

Fleiss' Kappa Evaluation Strategy for Choice of Reference MLLM

1 Overview

To select the reference model for APS question and golden triple generation, we conducted a pilot study using GPT-4o, Gemini 2.5 Pro, and Claude Sonnet 4.5. Three independent annotators evaluated the generated APS questions and golden triples according to predefined quality criteria. Inter-annotator agreement was measured using Fleiss' κ [1].

2 APS Evaluation Criteria

- **Correctness:** The APS sub-question should be factually consistent with the corresponding source problem and have a well-defined answer.
- **Reasoning Depth:** APS sub-questions should vary in complexity, requiring different numbers of supporting triplet hops to answer, thereby covering multiple levels of perceptual understanding.
- **Reasoning Leakage:** APS sub-questions should evaluate only perception and semantic understanding of the input, without requiring downstream reasoning or problem solving.

3 Golden Triple Evaluation Criteria

- **Structural Correctness:** Each knowledge graph triple should conform to a valid (*subject, predicate, object*) structure.
- **Factual Correctness:** The generated triples should accurately represent the information present in the source problem and provide sufficient information to answer the corresponding APS questions.
- **Minimality:** The triple set should provide a concise yet sufficient representation of the problem, avoiding redundant information that may increase token cost and hallucination risk.

4 Fleiss' Kappa

Fleiss' κ measures agreement among multiple annotators while correcting for agreement that may occur by chance.

Let:

- N denote the number of annotated items.
- n denote the number of annotators.
- k denote the number of annotation categories.
- n_{ij} denote the number of annotators assigning item i to category j .

4.1 Item-wise Agreement

For each item i , the agreement score is computed as

$$P_i = \frac{1}{n(n-1)} \sum_{j=1}^k n_{ij}(n_{ij} - 1). \quad (1)$$

where P_i represents the degree of agreement among annotators for item i .

4.2 Observed Agreement

The mean observed agreement across all items is

$$\bar{P} = \frac{1}{N} \sum_{i=1}^N P_i. \quad (2)$$

4.3 Category Proportions

The proportion of all ratings assigned to category j is

$$p_j = \frac{1}{Nn} \sum_{i=1}^N n_{ij}. \quad (3)$$

4.4 Expected Agreement

The agreement expected by chance is

$$\bar{P}_e = \sum_{j=1}^k p_j^2. \quad (4)$$

4.5 Fleiss' Kappa Score

The final Fleiss' κ score is computed as

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}. \quad (5)$$

A value of $\kappa = 1$ indicates perfect agreement, $\kappa = 0$ indicates chance-level agreement, and negative values indicate systematic disagreement among annotators.

5 Interpretation

Fleiss' κ Range	Interpretation
< 0.20	Slight Agreement
$0.21 - 0.40$	Fair Agreement
$0.41 - 0.60$	Moderate Agreement
$0.61 - 0.80$	Substantial Agreement
$0.81 - 1.00$	Near-Perfect Agreement

6 Reference Model Selection

Claude Sonnet 4.5 [4] was selected as the reference generator after a pilot study comparing GPT-4o [2], Gemini 2.5 Pro [3], and Claude Sonnet 4.5 [4] on 30 randomly sampled questions. Three independent annotators evaluated APS correctness, reasoning depth, reasoning leakage, structural correctness, factual correctness, and triple minimality. Claude achieved the highest inter-annotator agreement (Fleiss' $\kappa = 0.80$), indicating substantial to near-perfect agreement, and was therefore selected as the reference model for APS and golden triple generation.

References

- [1] Joseph L. Fleiss. Measuring Nominal Scale Agreement Among Many Raters. *Psychological Bulletin*, 76(5):378–382, 1971.
- [2] Josh Achiam et al. GPT-4 Technical Report. *arXiv preprint arXiv:2303.08774*, 2023.
- [3] Gheorghe Comanici et al. Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities. *arXiv preprint arXiv:2507.06261*, 2025.
- [4] Anthropic. Claude Sonnet 4.5. Available at: <https://www.anthropic.com/claude/sonnet>, 2025.